

The Adolescent Scar of Early-Life Disaster

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Tables and instrument appendix — draft for discussion

OVERVIEW

The 27 February 2010 earthquake, of magnitude 8.8, struck every child in this paper between birth and age four. Fourteen years later the ELPI survey returned to the same children and, for the first time, asked the adolescents themselves about their mental health. We find that the earthquake left a scar whose depth depends on the age at which it struck. Adolescents who lived through it at age two report fewer anxiety and depression symptoms at ages 15–18 than adolescents struck as infants, when attachment and stress regulation are forming, or at ages three and four, when children form their first lasting memories, a U-shaped profile whose trough is consistent with the window of infantile amnesia (Terr 1988; Bauer 2007; Alberini and Travaglia 2017). The caregivers of the same adolescents report no gradient at any age, and almost all of the self-reported gradient belongs to girls.

The magnitude is moderate. Measured with the PHQ-4, a four-question screener of low mood, loss of interest, nervousness and worry that the adolescents answer in private, exposure at age two lowers the score by 0.39 points on its 0-to-12 scale relative to exposure before age one, or 0.147 standard deviations, statistically significant at the 5 percent level (column 3 of Panel A in Table 3). The index averages 4.31 points in the sample, so the gap corresponds to 9 percent of the mean and to about a third of the 1.11-point distance between girls and boys, and the distance by mother's education is essentially zero. The anxiety half of the index moves the most, since the probability of a positive GAD-2 screen falls by 7.3 percentage points relative to a base of 27.3 percent.

Among girls the PHQ-4 gap doubles to 0.300 standard deviations and appears in both halves of the index, with a fall of 13.7 percentage points in the anxiety screen and of 11.7 in the depression screen, while among boys every estimate is close to zero and statistically insignificant (Table 13). Among caregivers the same design finds nothing at any age (Table 4), which is consistent with symptoms that adolescents feel and adults do not observe. Figure 1 draws the full age profile. The estimates are flat in a placebo on five outcomes fixed at birth (Table 6) and stable across every specification we tried (Table 7). Permutation of the affected zone across the 116 municipalities gives $p = 0.045$, and adjusting the four mental-health outcomes as a family leaves Romano–Wolf

p -values between 0.10 and 0.12, so we read the full-sample evidence as suggestive and the girls' result as the firm finding.

EMPIRICAL STRATEGY

Every adolescent of the 2024 wave lived through the earthquake as a child, and identification rests on the two things that vary across them, how hard their municipality was shaken and how old they were when it struck, with age grouped as 0–11 months (the omitted group), 12–23, 24–35 and 36–59. We compare the outcome gap between two exposure ages within the municipalities the earthquake hit against the same gap within the municipalities it did not hit, so municipality fixed effects keep every comparison local and birth-cohort fixed effects absorb everything a cohort shares, including its age at the survey. All tables estimate versions of

$$Y_{imc} = \alpha + \sum_a \beta_a (\text{Bin}_{ai} \times D_m) + \sum_a \gamma_a \text{Bin}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}, \quad (1)$$

where Y_{imc} is the outcome of adolescent i of birth cohort c in municipality m ; Bin_{ai} are the exposure-age groups; D_m equals one in the six regions where the official intensity study records destructive shaking (V to IX and the Metropolitan Region; Astroza et al. 2010), and the dose tables replace it with the standardized peak ground acceleration, the strength with which the ground actually shook in the municipality; ψ_m and θ_c are municipality-of-selection and birth-cohort fixed effects; X_i are household controls; and ε_{imc} is the error term. The coefficients of interest are the β_a , the effect of having lived the earthquake in an affected municipality at age a relative to the children exposed before age one. The main effects γ_a absorb the outcome differences across exposure-age groups that are common to affected and unaffected municipalities, differences that exist mechanically because a child who was older at the earthquake is also older at the 2024 interview, and β_a is therefore identified only from how that age profile differs between the two zones. Standard errors are clustered by municipality of selection, estimates use the 2024 sampling weights, and Table 7 adds permutation and wild-cluster inference for the preferred specification.

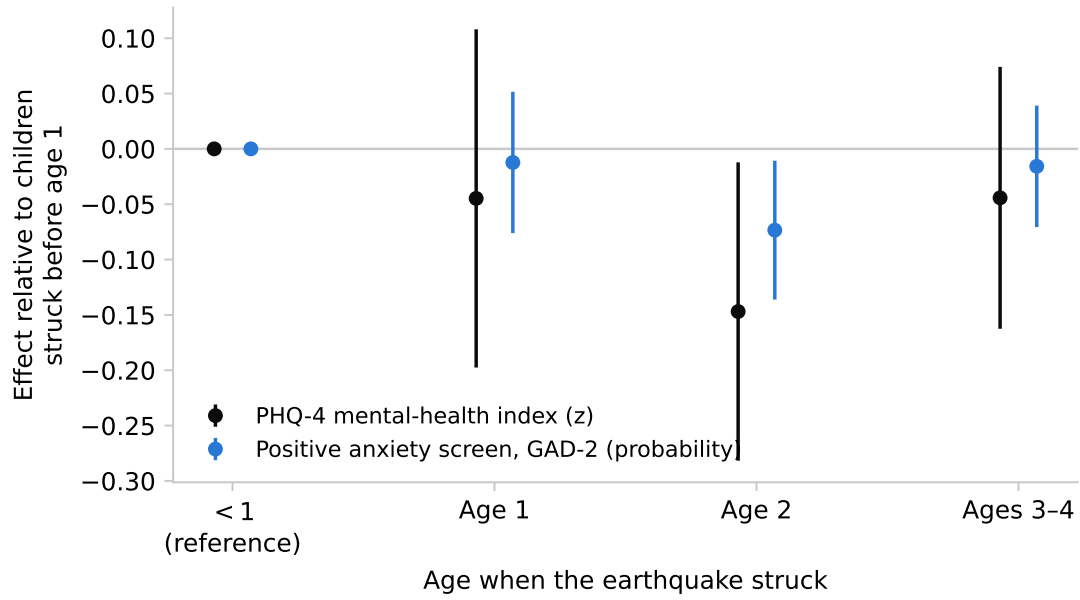


Figure 1. Estimates by Age at Exposure

Notes: Each marker is a β_a coefficient of equation (1) with the controls of column 3 of Table 3, with its 95 percent confidence interval. The reference group, drawn at zero by construction, are the children struck at 0–11 months — already born; children in utero during the earthquake are not in the 2024 wave, which follows only the cohort born 2006–2009. The PHQ-4 is in standard deviations and the GAD-2 screen in probability units. Standard errors clustered by municipality of selection in parentheses.

Table 1
Descriptive statistics.

	(1) Non-Earthquake	(2) Earthquake	(3) Difference
Girl	0.489 (0.500)	0.489 (0.500)	−0.000 (0.011)
Age at the earthquake (months)	26.930 (12.710)	27.098 (12.614)	0.169 (0.254)
Mother's age (2010)	28.980 (7.066)	29.676 (7.116)	0.695*** (0.196)
Mother's schooling (2010)	10.312 (3.182)	10.311 (3.180)	−0.002 (0.211)
HH size (2010)	4.865 (1.641)	4.948 (1.686)	0.083 (0.074)
Rural household (2010)	0.080 (0.271)	0.094 (0.292)	0.014 (0.035)
Children of the mother (2012)	2.165 (1.072)	2.155 (1.075)	−0.010 (0.029)
Prior mental health, mother (2012)	0.091 (0.288)	0.125 (0.331)	0.034*** (0.012)
Prior mental health, father (2012)	0.023 (0.150)	0.027 (0.161)	0.004 (0.004)
Prior mental health, relative (2012)	0.123 (0.328)	0.152 (0.359)	0.029* (0.017)
Observations	2,349	7,654	10,003

Notes: ELPI uses sampling weight at national level. The sample are the 10,003 adolescents of the 2024 wave, born January 2006–August 2009, aged 6 to 49 months on the day of the earthquake and 15 to 18 at the 2024 interview. Sex and ages are fixed at birth and the 2010 rows come from the pre-earthquake baseline; the 2012 rows are the family size and the lifetime conditions reported in the 2012 wave, the controls of the main specification, and are missing for adolescents without that interview. Mental health rows mark a reported diagnosis of depression, anxiety, bipolar disorder, schizophrenia, a personality disorder or an autism spectrum disorder. Earthquake municipalities are those of the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Column 3 is the difference from a regression on the earthquake indicator; standard errors clustered at the municipality of selection appear in parentheses—number of clusters: 116. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2

The impact of the earthquake on cognitive and non-cognitive development in the medium- and long-term.

	Medium-term effect (Gillmore) 2017 wave		Long-term effect 2024 wave	
	(1)	(2)	(3)	(4)
<i>Panel A: Peabody</i>				
Affected×Earthquake	−0.174** (0.0686)	−0.280*** (0.0866)	−0.240** (0.104)	−0.412** (0.180)
Observations	14,069	14,069	13,779	13,779
<i>Panel B: CBCL2</i>				
Affected×Earthquake	−0.129* (0.0763)	−0.220** (0.105)	0.084 (0.0884)	−0.066 (0.125)
Observations	11,568	11,568	11,038	11,038
Municipality FE	✓	✓	✓	✓
Birth Cohort FE	✓	✓	✓	✓
Child-Mother-HH control	✓	✓	✓	✓
Municipality linear trends	×	✓	×	✓

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score and CBCL is the Child Behavior Check List measure, oriented so that higher means fewer problems. Columns 1 and 2 reproduce Columns 3 and 4 of Table 3 in Gillmore (2026): children exposed between in utero and age four, with scores measured seven years later, at ages 7–11; our replication of his Column 3 on the public ELPI files gives −0.160 for Peabody and −0.103 for CBCL2. Columns 3 and 4 estimate the same two specifications on the 2024 wave: the same exposed children measured fourteen years later, at ages 15–18, against the comparison group conceived after the earthquake and measured in 2017; pooling the two waves adds survey-wave fixed effects, as in Gillmore’s short-term table. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145 in Columns 1 and 2 and 116 in Columns 3 and 4. The estimates use sampling weights. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers’ age and schooling, father present, number of household members, prior mental health condition for mother, father and relative. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3**Age at Exposure to the Earthquake and Self-Reported Mental Health at Ages 15–18**

	(1)	(2)	(3)	(4)
<i>Panel A. Mental health: PHQ-4 index, four questions on low mood, loss of interest, nervousness and worry (0–12, z)</i>				
Exposed at age 1	–0.051 (0.080)	–0.044 (0.078)	–0.045 (0.078)	–0.030 (0.079)
Exposed at age 2	–0.144** (0.072)	–0.153** (0.069)	–0.147** (0.069)	–0.120 (0.076)
Exposed at ages 3–4	–0.038 (0.064)	–0.048 (0.060)	–0.044 (0.060)	–0.011 (0.063)
<i>Panel B. The anxiety half of the index: positive GAD-2 screen (its two anxiety questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.014 (0.032)	–0.012 (0.032)	–0.012 (0.033)	–0.018 (0.035)
Exposed at age 2	–0.073** (0.032)	–0.076** (0.032)	–0.073** (0.032)	–0.064* (0.035)
Exposed at ages 3–4	–0.015 (0.027)	–0.018 (0.028)	–0.016 (0.028)	–0.015 (0.030)
<i>Panel C. The depression half of the index: positive PHQ-2 screen (its two depression questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.017 (0.039)	–0.014 (0.038)	–0.014 (0.038)	–0.020 (0.041)
Exposed at age 2	–0.046 (0.042)	–0.050 (0.041)	–0.048 (0.041)	–0.073 (0.045)
Exposed at ages 3–4	–0.008 (0.032)	–0.011 (0.031)	–0.010 (0.031)	–0.017 (0.035)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Observations	9,996	9,996	9,996	6,430
Selection municipalities (clusters)	116	116	116	67

Notes: Ordinary least squares estimates of equation (1) on the adolescents of the 2024 ELPI wave; seven of the 10,003 adolescents of Table 1 drop for incomplete PHQ-4 answers. Panels B and C disaggregate the Panel A index into its two pairs of questions, expressed as clinical screens; Appendix Table A1 lists every question verbatim. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. The affected zone comprises the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother's education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. The sample is the same in every panel; column 4 has 6,430 adolescents in 67 municipalities. Demanding inference for column 3 is reported in Table 7. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 4
Caregiver-Reported Outcomes

	(1)	(2)	(3)	(4)
<i>Panel A. CBCL internalizing score in 2024: caregiver checklist (T, z)</i>				
Exposed at age 1	−0.057 (0.099)	−0.047 (0.096)	−0.055 (0.095)	−0.024 (0.100)
Exposed at age 2	−0.121 (0.074)	−0.121 (0.075)	−0.109 (0.075)	−0.131 (0.084)
Exposed at ages 3–4	−0.086 (0.069)	−0.084 (0.068)	−0.077 (0.069)	−0.069 (0.074)
Observations	9,970	9,970	9,970	6,412
<i>Panel B. Within-child change in the same CBCL score, 2017 to 2024 (z by wave)</i>				
Exposed at age 1	−0.030 (0.131)	−0.011 (0.125)	−0.012 (0.124)	−0.020 (0.128)
Exposed at age 2	−0.009 (0.104)	−0.004 (0.102)	−0.002 (0.102)	−0.045 (0.110)
Exposed at ages 3–4	−0.040 (0.109)	−0.049 (0.106)	−0.050 (0.106)	−0.093 (0.112)
Observations	7,169	7,169	7,169	4,901
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Selection municipalities (clusters)	116	116	116	67

Notes: Same design and specifications as Table 3, with the caregiver’s report of the child as the outcome. The CBCL internalizing score is reported by the main caregiver, in T units, standardized within age. Panel B uses the change between the 2017 measurement (ages 8–12) and the 2024 measurement (ages 15–18) of the same child, each standardized within its wave and age; with two periods this is equivalent to a child fixed-effects estimate. Neither panel shows an exposure-age gradient: the scar of Table 3 appears only in what adolescents report in private. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother’s education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 5

The Gradient Under Continuous Seismic Doses

	Seismic dose of the municipality of selection			
	Affected zone (0/1)	PGA (z)	MMI (z)	−log dist. to epicenter (z)
	(1)	(2)	(3)	(4)
<i>Panel A. PHQ-4 score: sum of four depression–anxiety items (z)</i>				
Exposed at age 1	−0.045 (0.078)	−0.035 (0.033)	−0.028 (0.032)	−0.029 (0.029)
Exposed at age 2	−0.147** (0.069)	−0.068** (0.027)	−0.062*** (0.023)	−0.062** (0.025)
Exposed at ages 3–4	−0.044 (0.060)	−0.037 (0.027)	−0.039 (0.024)	−0.025 (0.023)
<i>Panel B. Positive GAD-2 screen: two anxiety items, subscore of 3 or more</i>				
Exposed at age 1	−0.012 (0.033)	−0.015 (0.014)	−0.013 (0.013)	−0.022* (0.013)
Exposed at age 2	−0.073** (0.032)	−0.039*** (0.015)	−0.029** (0.013)	−0.036*** (0.013)
Exposed at ages 3–4	−0.016 (0.028)	−0.016 (0.013)	−0.010 (0.010)	−0.014 (0.011)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes
Observations	9,996	9,996	9,996	9,996
Selection municipalities (clusters)	116	116	116	116

Notes: Estimates of equation (1) under four measures of the seismic dose, all with the controls of column 3 of Table 3. Column 1 repeats the official binary affected zone. Column 2 uses the peak ground acceleration: the strongest shaking of the ground during the earthquake, expressed as a share of the acceleration of gravity, measured at the centroid of the municipality of selection (from the ShakeMap of the U.S. Geological Survey) and standardized; column 3 an intensity scale from the same source; column 4 minus the log of the distance from the centroid to the epicenter. The continuous doses vary across the 116 municipalities, within affected regions, which addresses inference concerns with a regional treatment. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 6

Placebo: Outcomes Fixed at Birth, Before the Earthquake

	Birth weight (z) (1)	Birth length (z) (2)	Gestation weeks (z) (3)	Premature (0/1) (4)	Apgar 5' (z) (5)
Exposed at age 1	−0.013 (0.067)	−0.029 (0.074)	−0.023 (0.079)	0.017 (0.021)	−0.083 (0.087)
Exposed at age 2	−0.082 (0.080)	−0.074 (0.074)	−0.166** (0.069)	0.041* (0.024)	−0.043 (0.089)
Exposed at ages 3–4	0.017 (0.075)	0.014 (0.060)	−0.001 (0.077)	0.003 (0.021)	−0.013 (0.112)
Municipality fixed effects	Yes	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes	Yes
Observations	9,153	9,104	9,715	9,988	4,143

Notes: Each column reports one estimate of equation (1) on an outcome fixed at birth, one to four years *before* the earthquake for every cohort in the sample, as reported by the mother in 2010 (Appendix Table A5). The design should find nothing. Of fifteen coefficients, two are significant at the 10 percent level (1.5 expected by chance), and their sign — a worse birth endowment of the age-2 group in affected municipalities — would work *against* the trough of Table 3; the birth-endowment row of Table 7 confirms the result does not move. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 7
Robustness and Demanding Inference for the Main Result

	Dependent variable: PHQ-4 score (z)	
	Age 2 × affected (1)	Observations (2)
Preferred specification (column 3)	−0.147** (0.069)	9,996
Fixed effects only	−0.144** (0.072)	9,996
Excluding the Metropolitan Region	−0.120 (0.076)	6,430
Unweighted	−0.129* (0.071)	9,996
PHQ-4 in points (0–12), not standardized	−0.391** (0.183)	9,996
Balanced sample with a 2017 CBCL	−0.114 (0.082)	7,188
Controlling for the birth endowment	−0.144** (0.069)	9,996
Rewighted for attrition (IPW)	−0.149** (0.069)	9,996
Continuous dose: PGA (z) instead of the zone	−0.068** (0.027)	9,996
Dropping one affected region at a time (range)	[−0.172, −0.120]	
Municipality and birth-cohort fixed effects (all rows)	Yes	
<i>Demanding inference for the preferred specification (p-values)</i>		
Permutation of the affected zone	0.045	
Wild cluster bootstrap by region (15 regions)	0.132	

Notes: Each row re-estimates the preferred specification of Table 3, Panel A, changing one thing. Column 1 reports the coefficient on exposure at age two $\times$ affected zone. The IPW row reweights by the inverse of the estimated probability of remaining in the 2024 wave (Table 9). The permutation test reassigns the affected-zone status across the 116 municipalities of selection, keeping the number of treated municipalities, 2,000 times. The wild cluster bootstrap by region uses Webb weights and clusters at the level of the regional treatment. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

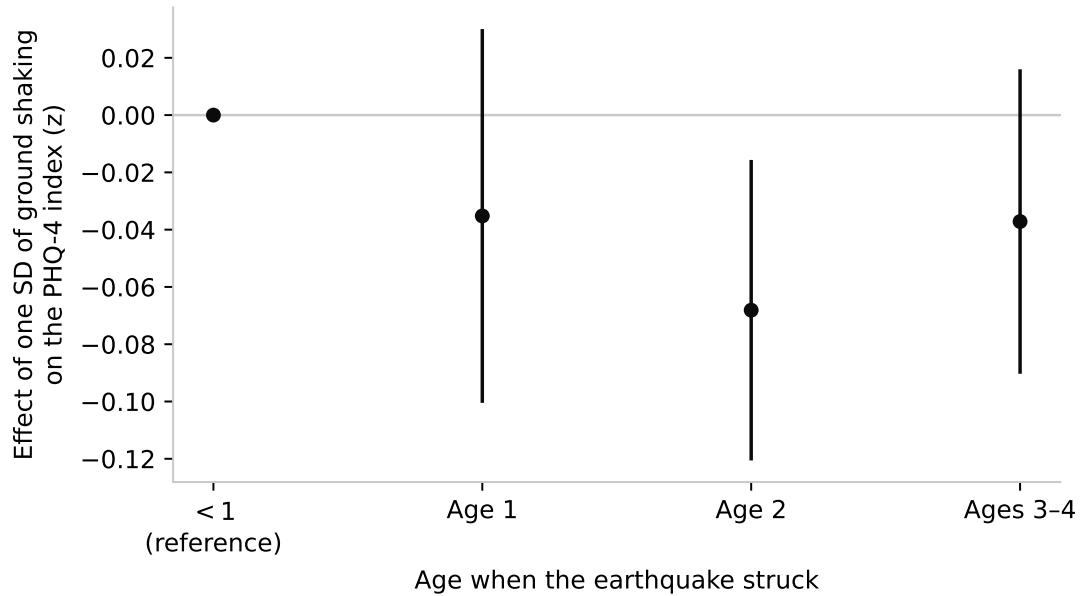


Figure 2. The Gradient Under the Continuous Seismic Dose

Notes: Each marker is the coefficient on the exposure-age group interacted with the standardized peak ground acceleration, how strongly the ground shook, at the centroid of the municipality of selection, with its 95 percent confidence interval; specification of column 3 of Table 3. The dose varies across the 116 municipalities, within affected regions. Standard errors clustered by municipality of selection in parentheses.

Table 8
Multiple-Hypothesis Adjustment Across Mental-Health Outcomes

	Age 2 $\times$ affected	
	Unadjusted p (1)	Romano–Wolf p (2)
PHQ-4 mental-health index (z)	0.033	0.120
Its depression half: PHQ-2 positive	0.247	0.299
Its anxiety half: GAD-2 positive	0.022	0.102
CBCL internalizing, caregiver (z)	0.147	0.299

Notes: Romano–Wolf stepdown p -values over the family of four mental-health outcomes, estimated with the preferred specification of Table 3. The adjustment uses a joint wild cluster bootstrap by municipality (Rademacher weights, 2,000 replications, the same sign flips for every outcome), for the age-two coefficient.

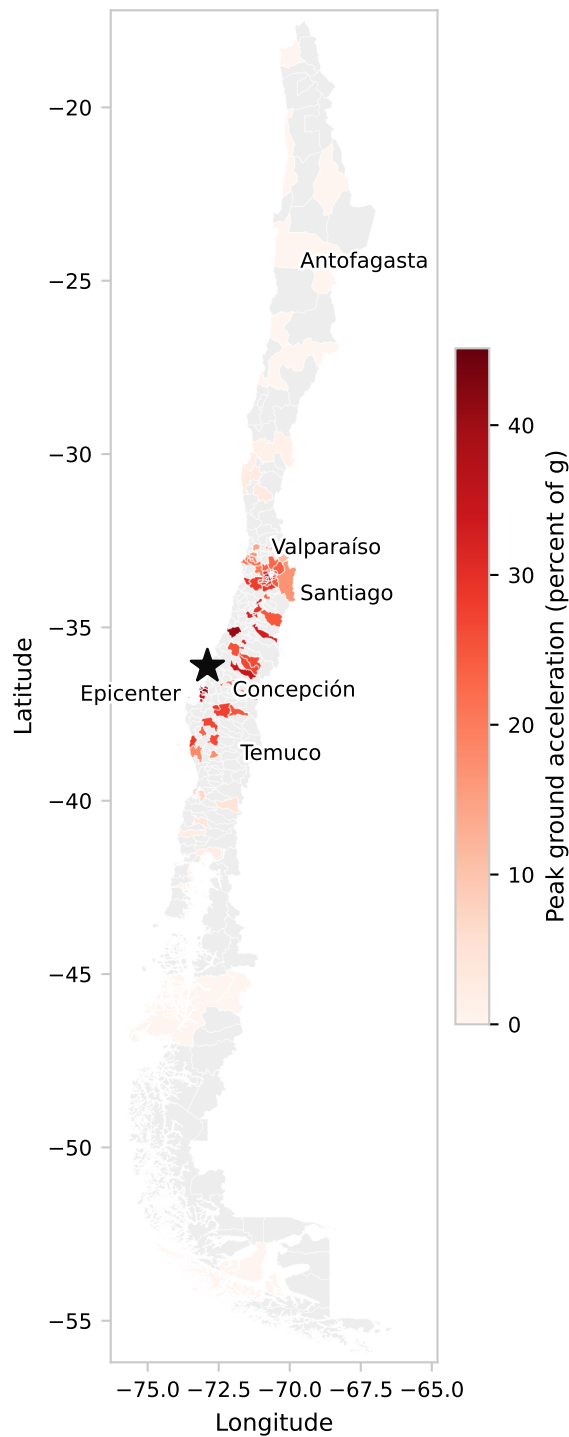


Figure 3. The Seismic Dose Across the Municipalities of Selection

Notes: Each colored municipality is one of the 116 municipalities of selection of the ELPI sample, colored by the peak ground acceleration, that is, how strongly the ground shook during the earthquake, expressed as a share of the acceleration of gravity (from the ShakeMap of the U.S. Geological Survey); the star marks the epicenter and light-gray municipalities are not in the sample. The map shows the two sources of identifying variation of Table 5, the affected zone and the continuous dose within it.

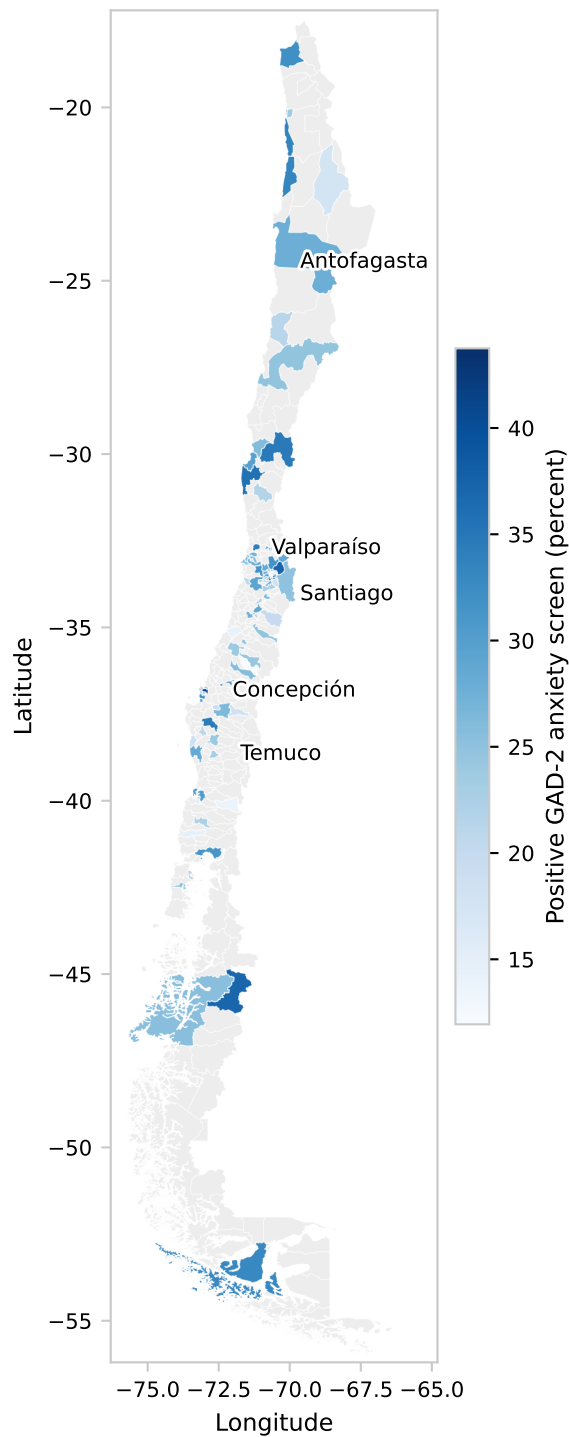


Figure 4. The Geography of Adolescent Anxiety in 2024

Notes: Each colored municipality is one of the 116 municipalities of selection, colored by the share of its adolescents with a positive GAD-2 anxiety screen in the 2024 wave; municipality shares range from 11.9 to 43.8 percent, and light-gray municipalities are not in the sample. The map is descriptive and unadjusted; the regressions of the paper compare cohorts within municipality, never municipalities against each other.

Table 9

Attrition from 2010 to 2024 by Exposure Age and Zone

	Pr(reinterviewed in 2024) (1)
Exposed at age 1 $\times$ affected	0.038 (0.029)
Exposed at age 2 $\times$ affected	0.028 (0.029)
Exposed at ages 3–4 $\times$ affected	0.038 (0.029)
Exposure-age and region fixed effects	Yes
Mother's education and age in 2010	Yes
Observations	14,855
Mean retention into 2024	67.3 percent

Notes: Linear probability model over the children of the 2010 baseline with a valid birth date. The outcome is an indicator for being reinterviewed in the 2024 wave. The small and insignificant coefficients show that the loss of the sample (32.7 percent) does not concentrate in any treatment-by-age cell; the fitted probabilities feed the IPW row of Table 7. The affected zone enters through the region of residence reported in 2010. Robust standard errors in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 10**Other Adolescent Outcomes I: Well-Being and Victimization**

	Resilience (z) (1)	Life satisf. (z) (2)	Poor health (z) (3)	Bullying (z) (4)	Cyber-vict. (0/1) (5)
Exposed at age 1	0.096 (0.076)	0.086* (0.050)	−0.086 (0.066)	0.028 (0.060)	−0.015 (0.029)
Exposed at age 2	0.037 (0.080)	0.098* (0.056)	−0.072 (0.077)	−0.047 (0.071)	−0.007 (0.023)
Exposed at ages 3–4	0.058 (0.070)	0.116* (0.062)	−0.054 (0.070)	0.037 (0.063)	−0.005 (0.025)
Municipality fixed effects	Yes	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes	Yes
Observations	9,996	9,996	9,996	9,176	9,985

Notes: Each column reports one estimate of equation (1) with the specification of column 3 of Table 3 on another outcome of the 2024 wave. Resilience is the mean of the six Brief Resilience Scale items; life satisfaction a single 1–7 item; poor health a single reversed item; bullying the mean of eight frequency items, one reverse-coded; cyber-victimization one if any of six items is answered yes. Appendix Table A3 lists every item and scale verbatim. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 11

Other Adolescent Outcomes II: Dating Violence and Substance Use

	Dating viol. (0/1) (1)	Tobacco (0/1) (2)	Alcohol (0/1) (3)	Cannabis (0/1) (4)
Exposed at age 1	0.025 (0.019)	0.012 (0.016)	−0.038 (0.032)	0.018 (0.017)
Exposed at age 2	0.005 (0.020)	−0.002 (0.015)	−0.037 (0.040)	0.014 (0.016)
Exposed at ages 3–4	0.028 (0.021)	−0.022 (0.024)	−0.078** (0.031)	0.013 (0.019)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes
Observations	7,101	9,995	9,995	9,995

Notes: Each column reports one estimate of equation (1) with the specification of column 3 of Table 3 on another outcome of the 2024 wave. Dating violence is one if any of three items is answered yes, among adolescents who ever dated; tobacco, alcohol and cannabis refer to the last 12 months, with alcohol one if any of four beverage items is answered yes. Appendix Table A3 lists every item and scale verbatim. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 12

Estimates by the Mother's Pre-Earthquake Mental Health

	PHQ-4 score (z)		Positive GAD-2 screen	
	Vulnerable mother (1)	Other mothers (2)	Vulnerable mother (3)	Other mothers (4)
Exposed at age 1	−0.036 (0.174)	−0.021 (0.080)	−0.022 (0.066)	−0.001 (0.036)
Exposed at age 2	−0.101 (0.129)	−0.124 (0.089)	−0.061 (0.063)	−0.062 (0.040)
Exposed at ages 3–4	−0.003 (0.127)	−0.017 (0.071)	−0.052 (0.063)	0.010 (0.031)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Observations	2,023	7,973	2,023	7,973

Notes: Split-sample estimates of the preferred specification of Table 3. A vulnerable mother reported, in 2010, a diagnosis of depression, anxiety or post-traumatic stress during the pregnancy of the child, a psychological or psychiatric referral, or diagnosed postpartum depression — all predating the earthquake by years (Appendix Table A4). The age-2 coefficient is nearly identical in the two groups, for both outcomes: the gradient is not a recomposition of pre-existing family mental-health burden. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 13
Heterogeneity of the Main Result

	Age 2 $\times$ affected (1)	Observations (2)
<i>Panel A. The PHQ-4 index (z)</i>		
Girls	−0.300*** (0.100)	4,961
Boys	−0.016 (0.098)	5,035
Mother with secondary education or less	−0.107 (0.074)	7,755
Mother with tertiary education	−0.201 (0.188)	2,053
Urban household in 2010	−0.139* (0.076)	9,005
Rural household in 2010	−0.238 (0.172)	991
<i>Panel B. The two halves of the index, girls versus boys</i>		
Girls, anxiety half (GAD-2 positive)	−0.137*** (0.047)	4,961
Boys, anxiety half (GAD-2 positive)	−0.026 (0.048)	5,035
Girls, depression half (PHQ-2 positive)	−0.117** (0.057)	4,961
Boys, depression half (PHQ-2 positive)	0.016 (0.045)	5,035
Municipality and birth-cohort fixed effects	Yes	

Notes: Each pair of rows estimates the specification of column 3 of Table 3 on the subsample named in the row. Panel A uses the PHQ-4 index; Panel B splits the index into its anxiety and depression halves, expressed as clinical screens, separately for girls and boys. The baseline level of symptoms is much higher among girls — the index averages 4.88 points for girls and 3.76 for boys. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

APPENDIX. INSTRUMENTS: EVERY ITEM AND ITS SCALE

Table A1
Mental-Health Instruments: Items and Scales

	Question, verbatim (1)	Response scale (2)
<i>Panel A. PHQ-4, answered by the adolescent in private (2024 wave)</i>		
d4_1	¿Te has sentido bajoneado(a), deprimido(a), irritable o desesperanzado(a)?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_2	¿Has sentido poco interés o placer al hacer las cosas?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_3	¿Te has sentido muy nervioso(a), angustiado(a) o con los nervios de punta?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_4	¿No has podido dejar de preocuparte o controlar la preocupación?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
<i>Panel B. Caregiver-reported behavior</i>		
cbc12	Child Behavior Checklist, internalizing scale, answered by the main caregiver about the adolescent; the survey releases the computed T score, not the copyrighted items.	T score, standardized in the analysis

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. The PHQ-4 score used in the paper is the sum of the four items rescaled to 0–3 each (0–12 in total) and standardized within age in years. The positive PHQ-2 (depression) and GAD-2 (anxiety) screens are the survey-provided indicators for a subscore of three or more on the corresponding pair of items.

Table A2
Well-Being Outcomes: Items and Scales

	Question, verbatim (1)	Response scale (2)
<i>Panel A. Resilience: Brief Resilience Scale (items 2, 4 and 6 reverse-coded)</i>		
d2_1	Tiendo a recuperarme rápidamente después de haber vivido situaciones difíciles . . .	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_2	Tengo facilidad para superar eventos estresantes	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_3	No me toma mucho tiempo recuperarme de un evento estresante	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_4	Es fácil para mí recuperarme cuando algo malo sucede	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_5	Usualmente supero situaciones difíciles con poca dificultad	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_6	Me toma poco tiempo recuperarme de las adversidades en mi vida	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
<i>Panel B. Life satisfaction and self-rated health</i>		
d3_1	Pensando en una escala de 1 a 7, donde 1 es nada satisfecho(a) y 7 es muy satisfecho(a), ¿qué tan satisfecho(a) estás con tu vida familiar?	1 = Nada Satisfecho(a); 2 = -; 3 = -; 4 = -; 5 = -; 6 = -; 7 = Muy Satisfecho(a)
d1	Considerando tu salud física, dirías que esta es:	1 = Muy mala; 2 = Mala; 3 = Regular; 4 = Buena; 5 = Muy buena

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. The resilience score is the mean of the six items after reverse-coding items 2, 4 and 6, standardized; life satisfaction and self-rated health are standardized, the latter with its sign reversed in the analysis so that higher values mean worse health.

Table A3**Victimization and Risk-Behavior Outcomes: Items and Scales**

	Question, verbatim (1)	Response scale (2)
<i>Panel A. School bullying (frequency scale; standardized mean; item g1_3 reverse-coded)</i>		
g1_1	Mis compañeros(as) se burlan de mí, me dicen sobrenombres	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_2	Me siento solo(a) en el curso	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_3	Lo paso bien con mis compañeros(as) de clase	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_4	Mis compañeros(as) son muy agresivos(as)	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_5	Mis compañeros(as) pelean mucho	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_6	A mis compañeros(as) les gusta hacer sufrir a los demás	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_7	La paso mal en la sala de clase	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_8	A mis compañeros(as) les gusta poner sobrenombres	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
<i>Panel B. Cyber-victimization, last 12 months (any item)</i>		
g2_1	¿En los últimos 12 meses has recibido mensajes ofensivos por medio de WhatsApp u otros servicios de mensajería?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_2	¿En los últimos 12 meses han publicado fotografías o videos vergonzosos de ti, a través de redes sociales?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_3	¿En los últimos 12 meses has recibido llamadas al celular insultantes o amenazantes?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_4	¿En los últimos 12 meses has recibido mensajes de correo electrónico insultantes o amenazantes?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_5	¿En los últimos 12 meses has leído o tenido conocimiento de conversaciones en grupos de chat en las que se expresan frases ofensivas sobre ti?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_6	¿En los últimos 12 meses has visitado o tenido conocimiento sobre páginas web donde se burlan de ti, o se descarga información personal sin tu permiso?	1 = Sí; 2 = No; 99 = Prefiero no responder
<i>Panel C. Dating violence, among adolescents who ever dated (any item)</i>		
g9_1	Violencia física	0 = No; 1 = Sí
g9_2	Violencia psicológica	0 = No; 1 = Sí
g9_3	Violencia sexual	0 = No; 1 = Sí
<i>Panel D. Substance use, last 12 months</i>		
g11	¿Has fumado cigarrillos de tabaco alguna vez en los últimos 12 meses?	1 = Sí; 2 = No
g12_1	Cerveza	1 = Sí; 2 = No
g12_2	Vino	1 = Sí; 2 = No
g12_3	Licores solos o mezclados con refrescos (pisco, vodka, ron, gin)	1 = Sí; 2 = No
g12_4	Cualquier otra bebida que contenga alcohol	1 = Sí; 2 = No
g13_1	Cannabis (Marihuana, hachís, creepy)	1 = Sí; 2 = No

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. Drank alcohol is one if any of the four beverage items is answered yes; cyber-victimization is one if any of the listed items is answered yes.

Table A4
Maternal Mental-Health Questions: Items and Scales

	Question, verbatim (1)	Response scale (2)
<i>Panel A. Reported by the mother in 2010, about the pregnancy and postpartum of the sampled child</i>		
g4a_1	Depresión. ¿Durante el embarazo le diagnosticaron alguna de las siguientes...	Marked = yes; not marked = no
g4a_3	Trastorno de ansiedad generalizada. ¿Durante el embarazo le diagnosticaron...	Marked = yes; not marked = no
g4a_7	Trastorno estrés postraumático. ¿Durante el embarazo le diagnosticaron a...	Marked = yes; not marked = no
g4b	¿Tuvo alguna derivación psicológica o psiquiátrica?	1 = sí; 2 = no; 9 = no sabe no responde
g19	Posterior al embarazo, ¿se le diagnosticó por un especialista, depresión po...	1 = sí; 2 = no; 9 = no sabe
<i>Panel B. Reported by the caregiver in 2012</i>		
b64	Últimamente, algún especialista, ¿le ha diagnosticado a Ud. depresión?	1 = sí; 2 = no; 8 = no responde
b55o_madre	Depresión. Le mencionaré una lista de enfermedades, ¿me podrí...	1 = sí; 2 = no; 9 = no sabe
b55p_madre	Ansiedad. Le mencionaré una lista de enfermedades, ¿me podría...	1 = sí; 2 = no; 9 = no sabe

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. A mother is vulnerable in Table 12 if any Panel A item is answered yes. The 2012 items enter the family mental-health controls of the main specification.

Table A5
Placebo Outcomes Fixed at Birth: Items and Scales

	Question, verbatim (1)	Response scale (2)
g24	¿Cuál fue el peso del(de la) niño(a) al nacer?	Open numeric answer (kilograms); special codes: 88 = dato impugnado; 99 = no sabe no responde
g23	¿Cuál fue la talla, altura del niño(a) al nacer?	Open numeric answer (centimeters); special codes: 88 = dato impugnado; 99 = no sabe no responde
g20	¿Cuántas semanas de embarazo tenía Ud. cuando nació el(la) niño(a)?	Open numeric answer (weeks); special codes: 88 = dato impugnado; 99 = no sabe no responde
g17_6	Parto prematuro (antes de las 37 semanas). Al momento del nacimiento, ¿ex...	1 = sí; 2 = no; 9 = no sabe no responde
g25b1	b1)Medición 2 (5 Minutos). ¿Cuál fue su puntuación en el Test de Apgar?	Open numeric answer (score 0–10); special codes: 99 = no sabe no responde

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. Cleaning: birth weight kept between 0.5 and 6.5 kilograms, length between 30 and 65 centimeters, gestation between 24 and 45 weeks, Apgar between 0 and 10; the survey codes 99 for a forgotten answer, treated as missing. All five outcomes are fixed at birth, one to four years before the earthquake for every cohort of the analysis sample.